Creating Emojis using Generative AI

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***Abstract*—This study investigates the capability of generative artificial intelligence (GAI), specifically text-to-image generation (TTIG) models like Stable Diffusion XL (SDXL) and Generative Pre-trained Transformer 4 Omni (GPT-4o), to create emotionally expressive emojis comparable to human designs. Using a positivist, quantitative research approach, AI-generated emojis were evaluated against Microsoft, Apple, and Google emojis using metrics like CLIPScore, VQAScore, LPIPS, and LAION-Aesthetics V2. The findings show that SDXL excelled in semantic alignment and GPT-4o in stylistic consistency. Both AI-generated sets surpassed human emojis in aesthetic appeal. However, human evaluators could still distinguish AI-generated emojis. Future research should incorporate diverse human feedback to better contextualise emotional interpretations, evaluate in real-world communication settings, and explore the impact of creating custom emojis.**

***Index Terms*—Emojis, Generative AI, ComfyUI, Stable Diffusion XL, GPT-4o**

I. Introduction

A. Description of Theme and Topic Rationale

Generative Artificial Intelligence (GAI) refers to technologies capable of generating new and meaningful content, such as text, images, or audio, from training data [1]. One popular technology in GAI is Text-to-image generation (TTIG), which creates images from text [2]. A promising application of TTIG is the generation of emojis. graphic symbols that represent facial expressions, emotions, abstract concepts, gestures, objects, animals, activities, and more [3]. Emojis are a form of non-verbal communication in computer-mediated communication (CMC), helping users express emotions and intentions in digital conversations. Unlike emoticons, which use punctuation marks to depict expressions, emojis are standardised visual symbols encoded in Unicode, allowing for universal use across digital platforms [3]. However, current emojis are limited by predefined designs, which may not always capture nuanced emotions or diverse cultural expressions. AI-generated emojis have the potential to address these limitations by creating dynamically generated symbols that better align with user intent.

B. Positioning and Research Onion

This study will use a positivist approach as it evaluates the AI-generated emojis through quantitative metrics for emotion accuracy and consistency. A positivist approach is appropriate as the outcome of the study is independent of influence from the researcher, and the evaluation of the generation of emojis is quantifiable using AI tools for accuracy score and similarity ratings. The research will follow a deductive approach, aiming to test predefined hypotheses, with an experimental strategy, comparing the generated emojis against human-designed emoji datasets such as OpenMoji using similarity ratings, along with accuracy scores with AI tools. This allows the researcher to start from a general theoretical framework and gather specific conclusions through experiments with controllable variables and observable cause-and-effect relationships.

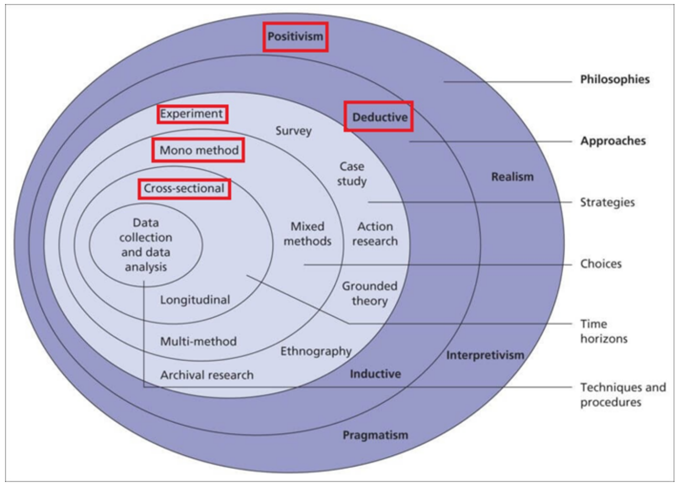

Fig. 1. Research Onion

The analysis will be conducted in a cross-sectional manner, where data is collected at a single point in time, after the emojis have been generated and prepared for evaluation. Finally, a mono-method quantitative strategy will be implemented, as the hypothesis and research questions can be effectively answered using quantitative methods alone, providing reliable measurements of variables such as accuracy and similarity.

C. Background to this research theme

Early approaches of TTIG, such as Generative Adversarial Networks (GANs), played an important role in image generation, however, they were limited in the ability to directly generate images from text prompts [4]. This led to the development of Diffusion Models, utilising Markov chains and variation inference to simulate data diffusion in a latent space, then reverse the process to generate high-quality images [1]. Moreover, the introduction of Contrastive Language-Image Pretraining (CLIP) improved models with understanding and aligning text with images [5], leading to more sophisticated TTIG models. Certain tools and techniques will be used in this research, such as ComfyUI, a node-based graphical interface that facilitates the creation of workflows for conversions and image post-processing [4].

D. Research Aim & Purpose Statement

Given the increasing reliance on digital communication, this study aims to explore the potential of TTIGs in creating emotionally accurate emojis. Generative Pre-trained Transformer 4 Omni (GPT-4o) and a ComfyUI workflow using Stable Diffusion XL (SDXL) will be used to generate different emojis with different emotions. Given this aim, the following hypothesis was formed: *AI-generated emojis can successfully convey the intended emotions at a similar level to human-designed emojis.*

The purpose of this experimental study is to test the theory that GAI can produce emotionally expressive emojis that predict that the source of emoji design relates to the accuracy of the emotional expression, aesthetic and stylistic consistency [6]. This study aims to evaluate using computational tools. The independent variable, the source of emoji design, will be defined as the method of how the emoji was created, whether it was human-designed or AI-generated [6]. The dependent variables will be defined as; the accuracy of the emotional expression, the aesthetic of the emoji, and the stylistic consistency [6].

II. Literature Review

This section reviews the methodologies adopted in recent studies related to AI-generated imagery and emoji synthesis, with a focus on their research philosophy, methodological approach, and evaluation strategies. To provide a structured and comparative lens, this review draws upon the research onion model proposed in [7], as well as the framework on research design proposed in [6].

All of the following five studies adopt a positivist stance, prioritising objective, quantifiable outcomes, and follow a deductive reasoning process by testing hypotheses derived from existing theories. Furthermore, each study employs an experimental design, whether through psychological experiments in the case of [8], or through controlled GAN training environments as seen in [9], [10], [11], and [12]. Furthermore, a cross-sectional time horizon is used throughout all studies, with data collection and analysis taking place at a single point in time.

A. Methodologies of Similiar Studies

[8] aims to investigate whether AI-generated faces are distinguishable from real ones and examines trustworthiness perceptions. Three experiments were conducted: one tested unaided classification of real and synthetic faces with 315 participants, another with 219 participants which were trained and received feedback, and finally 223 participants were tasked with rating trustworthiness us a seven-point scale. Participants were recruited from Amazon Mechanical Turk's Master Worker pool, with data integrity being enhanced through quality control mechanisms, such as catch trials and performance incentives. Statistical procedures such as logistic regression and t-tests were employed to evaluate hypothesis-driven questions [8]. The research follows a positivist, mono-method quantitative design, emphasizing objective measurable data such as classification accuracy and trust ratings [8]. For each experiment, specific variables were manipulated while maintaining control over others to assess the effects on accuracy or perception [8]. Future research should consider including more culturally diverse participants to enhance external validity. Additionally, the study assumes a linear interpretation of ordinal trustworthiness ratings, enabling calculation of means and t-tests, future work might also explore non-parametric analyses or alternative scale designs fully reflect subjective impressions.

[10], introduces EAE-GAN, a GAN trained on 276 Unicode face emojis augmented to 3,036 samples, to generate facial emojis conditioned on an 8-dimensional emotion vector representing Plutchik's eight basic emotions. The research employs a mono-method quantitative design, evaluating and comparing EAE-GAN to other models such as PGGAN, Style-ALAE, and StyleGAN using objective outcomes. Each model generated 10,000 emojis under the same environment, with Fréchet Inception Distance (FID), Learned Perceptual Image Patch Similarity (LPIPS), and generation time being recorded. FID measures the quality of generated images by calculating the distance between the distribution of generated and real images, where a lower score indicates higher visual realism. On the other hand, LPIPS evaluates the diversity of generated images by calculating the distance between visual representations, with higher scores indicating greater perceptual diversity [10]. While human evaluation is conducted, it is constructed within a structured survey format that converts subjective judgments into numerical scores. A survey of 45 educated participants was conducted to assess emotional accuracy, of which 35 responses were retained after filtering for completeness and consistency. Participants rated 30 AI-generated emojis, by the perceived intensity of eight basic emotions on a scale of 0 to 100. The authors employed one-sample t-tests to compare input emotion vectors against participant responses, and Pearson correlation coefficients to assess the alignment between intended and perceived emotional content [10]. The sampling methods and demographic details used for the survey were not reported or given a specific justification. The study includes a visual case study to illustrate how EAE-GAN can interpolate emotional expressions between two emoji states. This involved adjusting the emotion vector across ten linear steps between two existing emojis, visually showcasing the ability of the model to capture subtle emotional gradations [10].

[9] investigates the feasibility of generating realistic cartoon emojis using a Deep Convolutional Generative Adversarial

Network (DCGAN). The research employs a primarily quantitative mono-method approach, rooted in data-driven techniques such as loss calculations and epoch values. However, evaluation includes a qualitative component, as performance is also assessed through visual inspection of generated images, which involves subjective interpretation [9]. Training was conducted under controlled conditions, using the Cartoon Set dataset, consisting of 10,000 emoji images, with the model evaluated over 20 epochs. The model includes a generator that creates images from noise, and a discriminator that classifies images as real or fake. Losses were calculated using the Binary Cross-Entropy loss function (BCELoss), and the Adam optimizer was used for backpropagation. Evaluation was performed through visual inspection and loss graphs [9]. While effective in demonstrating technical feasibility, the inclusion of standardised metrics such as FID, LPIPS or user-based evaluations in future studies could further strengthen the assessment by providing more objective and user-centred insights. The study also focuses exclusively on the DCGAN model without testing alternative architectures, limiting broader theoretical insights.

[11] proposes another DCGAN solution, optimizing a model to generate new emojis by training the generator and discriminator with a dataset of existing emojis from platforms like Facebook, Apple, and Google, pre-processed to 64×64 pixels to facilitate faster training. The results are evaluated at various epoch stages such as 100, 500, and 1000 epochs, using objective criteria such as the loss function and image clarity. The images are compared to real emojis, and the losses of the generator and discriminator are monitored throughout the training process [11]. The DCGAN model is implemented with the Adam optimisation algorithm, and BCELoss is used to train the discriminator. The study adopts a rigorous and reproducible mono-method quantitative design. [11]. However, the study could benefit from a more detailed critique of the dataset biases, such as variation in emojis styles potentially influencing the model's ability to generalise across platforms. Additionally, the pre-processing decisions, such as resizing images to 64×64 pixels, are not critically evaluated in terms of their potential impact on the quality of the generated emojis.

[12] investigates the use of Conditional Generative Adversarial Networks (cGANs) to generate emojis from text inputs. The model builds upon DCGAN with enhancements such as a heuristic objective function and batch-label training. A pre-trained word2vec model is used to convert text inputs into embeddings, which are then fed into the generator alongside noise vectors [12]. The EmotiGAN model was trained 2,000 Unicode emojis using both standard random-label batching and a novel same-label batching with minibatch discrimination. Model performance was evaluated using visual inspection and comparing inception scores, measuring quality and diversity, and is calcluated using the discriminator was converted into a classifier with a softmax output layer [12]. However, reliance on the modified inception score, could risk bias, as the discriminator was trained on the same dataset, potentially leading to artificially inflated performance scores. In terms of methodological choices, the study adopts a mono-method quantitative approach, while visual outputs are presented for qualitative judgment, no formal analysis is performed, reflecting a strong preference for quantitative rigour [12].

B. Comparisons of Studies

Across the reviewed studies, a consistent research philosophy and methodological approach are evident as seen in Table I and mentioned at the start of the section.

Despite these overarching similarities, differences emerge in the specific research designs and evaluation methods employed, as seen in Table II. While [8] employs human participants to assess face classification and trustworthiness, the other studies primarily rely on computational evaluations. For instance, [9] and [11] focus on GAN loss values and visual inspection, without standardised metrics like FID. In contrast, [10] strengthens their evaluation with both objective measures of FID, LPIPS and structured human surveys, offering a more holistic assessment of model performance. Meanwhile, [12] emphasises inception scores without formal qualitative analysis, reflecting a strong quantitative bias. These methodological nuances suggest that while the field uses similar philosophies, significant variation exists in how evidence is gathered and interpreted.

TABLE I
COMPARATIVE OVERVIEW OF STUDIES

Study	Focus	Philosophy & Approach	Strategy & Time Horizon
[8]	AI-synthesised faces: trustworthiness, realism	Positivist, Deductive	Experimental (human evaluation), Cross-sectional
[9]	Cartoon emoji generation (DCGAN)	Positivist, Deductive	Experimental (GAN training), Cross-sectional
[10]	Emotion-aware emoji generation (EAE-GAN)	Positivist, Deductive	Experimental (GAN + survey), Cross-sectional
[11]	Emoji generation (DCGAN from multiple platforms)	Positivist, Deductive	Experimental (GAN training), Cross-sectional
[12]	Text-to-emoji generation (cGAN with word embeddings)	Positivist, Deductive	Experimental (GAN training), Cross-sectional

TABLE II
RESEARCH DESIGN AND EVALUATION TOOLS

Study	Research Design	Data Collection	Evaluation Tools
[8]	Mono-method quantitative	Amazon MTurk, face classification, trustworthiness ratings	Logistic regression, t-tests, catch trials, incentives
[9]	Mono-method quantitative + visual inspection	DCGAN training on Cartoon Set dataset	Loss graphs (BCELoss, Adam optimizer), visual inspection
[10]	Mono-method quantitative + structured human evaluation	DCGAN training, emotion ratings from survey participants	FID, LPIPS, t-tests, Pearson correlations
[11]	Mono-method quantitative	DCGAN training on emoji datasets from multiple platforms	Loss analysis, checking visual quality
[12]	Mono-method quantitative	cGAN outputs, Word2Vec embeddings on Unicode emoji dataset	Modified inception scores, visual outputs

III. RESEARCH METHODOLOGY

This study aims to explore the potential of TTIGs in creating emotionally accurate emojis.

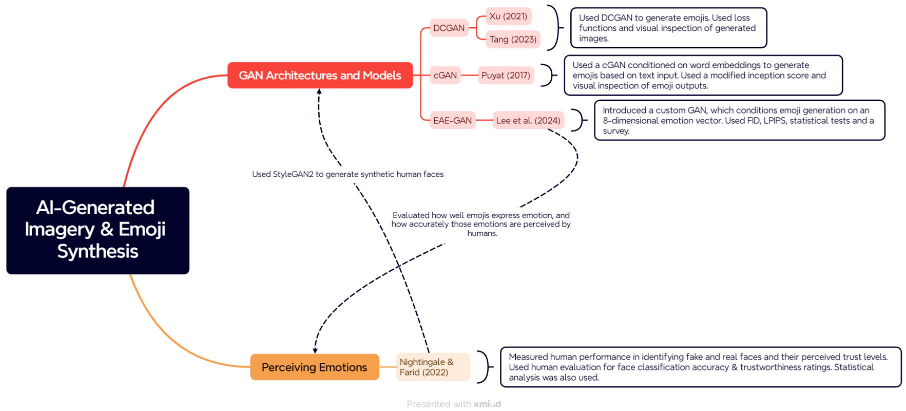

Fig. 2. Literature Map

Based on this aim, the following research questions were formed:

1) **RQ1:** To what extent can AI-generated emojis accurately convey intended emotions compared to human-designed emojis?
2) **RQ2:** How do AI-generated emojis compare to human-designed emojis in terms of aesthetic appeal and stylistic consistency?

To answer these research questions, the following objectives were constructed:

1) Test GPT-4o and a ComfyUI workflow with SDXL to generate emojis representing different emotions.
2) Collect a dataset of AI-generated emojis and comparable human-designed emojis.
3) Apply computational evaluation tools to objectively measure emotional expression accuracy, aesthetic quality, and stylistic consistency.
4) Statistically compare the performance of AI-generated and human-designed emojis based on the collected computational scores.

A. Research Design

As stated in the introduction, this study adopts a positivist philosophy. A positivist stance is appropriate because the research aims to objectively measure and evaluate the emotional accuracy, aesthetic quality, and stylistic consistency of AI-generated emojis using quantifiable computational metrics. The evaluation outcome is independent of the researcher's subjective interpretation, aligning with the principles of positivism.

A quantitative, mono-method, and deductive research design is employed as the study proposes a predefined hypothesis, that *AI-generated emojis can convey intended emotions at a similar level to human-designed emojis*, and evaluates this hypothesis through experimental and quantitative measurements. The data will be collected at a single point in time, following a cross-sectional time horizon after the emojis have been generated and prepared for evaluation. Given that this study aims to test a specific hypothesis through controlled experiments and computational metrics, the chosen research design is justified. The research focuses on observable cause-and-effect relationships between the source of emoji design and emotional expression outcomes, using objective evaluation tools. A mono-method quantitative strategy ensures the hypothesis and research questions are answered with reliable, replicable measurements without the need for qualitative exploration. The choice of research design is also supported by the methodologies reviewed in the literature. Studies investigating AI-generated faces, GAN-based emoji synthesis, and emotion-aware generative models similarly adopted positivist philosophies, deductive frameworks, quantitative strategies, and cross-sectional designs. Therefore, this research design is both appropriate for the aim of the study and consistent with established practices in the field.

B. Environment and Technologies Used

SDXL is selected because it offers a wide range of checkpoints and LoRA (Low-Rank Adaptation) models, allowing the choice of various art styles, especially those trained on emoji sets. Emoji generation workflows are built using ComfyUI, a node-based interface for Stable Diffusion [4], with support for custom nodes. For example, the `rembg` Python package[1] will be utilised within ComfyUI to remove backgrounds from generated emoji images automatically. A ChatGPT chat will be used to generate the images using GPT-4o. Evaluation will be automated using Python scripts on a local machine with a GTX 1070 GPU with CUDA support to ensure efficient performance during batch testing. Results will be using Microsoft Excel analyzed in IBM SPSS Statistics.

C. Experiment Design

The experimental dataset will consist of four distinct sets of emojis:

1) 10 AI-generated emojis created using SDXL.
2) 10 AI-generated emojis created using GPT-4o.

[1]https://github.com/danielgatis/rembg

3) 10 Microsoft emojis sourced from the Microsoft Emoji List[2].
4) 10 Google emojis sourced from the Google Emoji List[3].
5) 10 Apple emojis sourced from the Apple Emoji List[4].

Each emoji will represent a predefined emotional category, ensuring direct comparability across all groups. In designing the experiment, the selection of emotional categories was guided by [13], which identifies six fundamental emotions: anger, surprise, disgust, enjoyment, fear, and sadness. Additional emotions such as love, confusion, pride, and embarrassment were included. These emotions introduce complexity and subtlety, allowing for a more comprehensive assessment of the ability of the AI to convey nuanced emotional expressions. Table III showcases the emotion and the official Unicode name of the emojis that will be used in the experiments.

Emotion	Official Unicode Name
Enjoyment	Slightly Smiling Face
Sad	Slightly Frowning Face
Angry	Angry Face
Surprised	Astonished Face
Fearful	Fearful Face
Disgusted	Nauseated Face
Love	Smiling Face with Heart-Eyes
Confused	Confused Face
Proud	Smirking Face
Embarrassed	Flushed Face

TABLE III
SELECTED EMOTIONS AND THEIR CORRESPONDING OFFICIAL UNICODE EMOJI NAMES

For evaluation, several computational metrics will be employed, such as LPIPS, as identified and reviewed in the related literature. LPIPS will assess the perceptual similarity between the AI-generated emoji and each of the corresponding human-designed emojis. Additionally, LPIPS will also be used to calculate pairwise perceptual similarity between all combinations of the AI-generated emojis themselves. The mean LPIPS score will then be computed to evaluate the stylistic consistency within the AI-generated emoji set.

CLIPScore measures the cosine similarity between the image and corresponding text embeddings [14], and is recognised for achieving a high correlation with human judgment [14], [15]. However, CLIPScore has known limitations when prompts involve complex attributes, relationships, or varied compositions, leading to the development of VQAScore [15]. VQAScore applies a visual-question-answering model to more accurately assess the alignment between the visual output and the intended description. CLIPScore and VQAScore will be computed for each emoji by pairing the image with its intended emotional label as a text prompt, quantifying how well the visual content matches the emotional intent. Text emebeddings will be gathered and slightly modified from emoji descriptions in Emojipedia[5]. Additionally, aesthetic quality will be evaluated using the LAION-Aesthetics V2 model[6], which predicts how highly people are likely to rate the visual appeal of an image. In addition to these computational metrics, a user study in the form of a Turing-style test will be conducted with a sample size of 25 participants, the size of an average classroom, particularly sampling students from Malta College of Arts, Science and Technology (MCAST). Participants will be presented with 10 sets of 5 emojis each, where 2 emojis in each set are AI-generated. Participants will be instructed to identify which emojis they believe were AI-generated. This evaluation aims to measure how distinguishable the AI-generated emojis are from human-designed ones.

This experiment design directly supports the research objectives by enabling a structured comparison between AI-generated and human-designed emojis. The selected emotional categories ensure coverage of both basic and complex emotions, while the chosen computational metrics provide objective, quantifiable evaluation across emotional accuracy, aesthetic appeal, and stylistic consistency. The use of IBM SPSS Statistics for statistical analysis further reinforces the quantitative and positivist approaches of the study, allowing for rigorous and replicable testing of the proposed hypothesis.

D. Reflections on Validity, Reliability, and Generalizability

This study carefully considered validity, reliability, and generalizability during the experiment design phase.

Internal validity was strengthened by using clearly defined emotional categories and standardised computational evaluation metrics such as LPIPS, CLIPScore, VQAScore, and LAION-Aesthetics V2. These tools are grounded in established theory and literature, ensuring the results are accurately measured. By directly pairing emojis with their intended emotional labels, the evaluation approach ensures construct validity.

To ensure reliability, all emoji generation and evaluation procedures were standardised and performed under consistent conditions within the same hardware and software environment. Python scripts and ComfyUI workflows were used to automate the process, minimising variability. The use of objective computational metrics rather than subjective human ratings further enhances the consistency and reproducibility of the results.

The generalizability of the results is somewhat limited as the study focuses on a specific set of emotions and comparisons to particular human-designed emoji sets. Therefore, the conclusions are most applicable within the context of emoji-style visual communication and may not directly extend to other forms of AI-generated art or different cultural interpretations of emotional expressions.

E. Ethical Considerations

In the study, human-designed emojis from Microsoft, Google, and Apple are used for comparison purposes, falling under fair use for academic, non-commercial use.

[2]https://emojipedia.org/microsoft
[3]https://emojipedia.org/google
[4]https://emojipedia.org/apple
[5]https://emojipedia.org/
[6]https://laion.ai/blog/laion-aesthetics/

This study utilises checkpoints and LoRAs, which can be trained on copyrighted images. The use of these copyrighted images without explicit permission introduces potential intellectual property concerns. While the application of this model is confined to non-commercial, academic research, it is important to acknowledge the ethical implications of using training data that may infringe upon existing copyrights. The study remains vigilant in evaluating the outputs to ensure they do not replicate proprietary designs, thereby mitigating potential legal and ethical issues.

Additionally, this study acknowledges the ethical challenges associated with AI-generated imagery, particularly regarding bias and authenticity [16]. Since the models are trained on vast and often non-transparent datasets, there is a risk that the generated emojis may unintentionally perpetuate biases or stereotypical representations of emotions. During evaluation, care will be taken to remain critical of potential biases, and all generated outputs will be assessed through objective computational tools.

The study also maintains a clear distinction between AI-generated and human-designed emojis to preserve transparency for any future interpretations of the work. No human participants are involved, minimising concerns around consent or data privacy.

However, this project recognises that AI-generated visual content can contribute to changing traditional business models by reducing reliance on human artists and designers [16]. As such, the research also remains conscious of the broader societal implications of normalising AI-created emotional imagery.

IV. Findings & Discussion of Results

Each source was evaluated using the following: Each source was evaluated using the following:

1) LAION-Aesthetics V2 to determine an aesthetic score out of 10. Scores were relatively low across all sources, suggesting model limitations in measuring iconographic images such as emojis.
2) CLIPScore to measure the semantic alignment between the image and a text embedding, with scores clustering around 30%, as it struggles with complex attributes such as emotions, as mentioned previously in [15]. Therefore, this score should not be interpreted as a limit on model accuracy.
3) VQAScore to measure a more accurate semantic alignment between the image and text embedding through the use of question-answering. The score is normalised between 0 and 1.
4) LPIPS to measure perceptual dissimilarity between images. The score is normalised between 0 and 1.

All score should be interpreted in comparison to the human-designed sources. Table IV presents the average scores of five emoji datasets, two AI-generated and three human-authored, across the previously mentioned metrics. GPT-4o achieved the highest average aesthetic score with 5.092, followed by Apple with 5.029, and SDXL with 5.006; meanwhile, Microsoft scored the lowest with 4.829. In terms of CLIPScore, SDXL

TABLE IV
Average Scores Across Sources

Source	Aesthetic Score	CLIPScore	VQAScore	LPIPS Across Set
GPT-4o	5.092	29.207	0.550	0.297
SDXL	5.006	30.692	0.816	0.428
Apple	5.029	29.159	0.767	0.390
Google	4.919	28.964	0.644	0.259
Microsoft	4.829	28.726	0.558	0.319

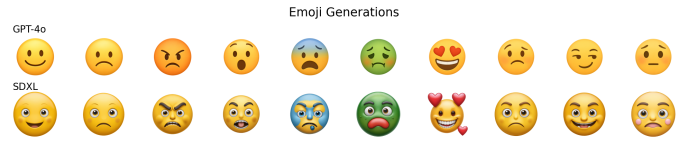

Fig. 3. Emoji Generation Samples

scored the highest with 30.692, followed by GPT-4o with 29.207. Microsoft scored the lowest with 28.726. Regarding the VQAScore, SDXL scored the highest with 0.816, and GPT-4o scored the lowest with 0.550. Finally, for the LPIPS across the set, with lower values indicating greater similarity within the set, Google achieved the lowest score with 0.259, followed by GPT-4o with 0.297, suggesting high perceptual consistency. In contrast, SDXL has the highest LPIPS, indicating a higher variation across outputs. To assist in answering the research question, analyses were conducted within IBM SPSS with the collected raw data.

V. Human-Authored & AI-Generated Comparison

An independent t-test was conducted to determine the significance of the CLIPScore, VQAScore and aesthetic score on the AI-generated dataset of 20 images comprising images generated by SDXL and GPT-4o in comparison to the human-authored dataset, which includes 30 emojis from Apple, Microsoft and Google. The results of the independent t-test are

TABLE V
Independent Samples t-Test Results: AI vs Human Models

Measure	F	Sig.	t	df	One-sided p	Two-sided p	Mean Diff.
Aesthetic Score	0.687	0.411	2.353	48	0.011	0.023	0.124
VQAScore	1.628	0.208	0.594	48	0.278	0.555	0.0267
CLIPScore	0.020	0.888	1.912	48	0.031	0.062	0.999

shown in table V. AI-generated emojis scored significantly higher in aesthetic ratings compared to human-authored ones with a two-sided p of 0.023, indicating that LAION-Aesthetics V2 perceived AI outputs as more visually appealing.

In contrast, the difference in VQAScore was not statistically significant with a two-sided p of 0.555, suggesting that both human and AI emojis were similarly effective in conveying their intended semantic content. The CLIPScore showed a marginal difference with a two-sided p of 0.062, with AI-generated emojis aligning slightly better with the emoji description.

VI. SDXL & GPT-4o Comparison

An independent t-test was conducted between the scores of SDXL and GPT-4o, with the results shown in VI. SDXL

TABLE VI
INDEPENDENT SAMPLES T-TEST RESULTS: SDXL VS GPT-4O

Measure	F	Sig.	t	df	One-sided p	Two-sided p	Mean Diff.
Aesthetic Score	0.140	0.712	0.890	18	0.193	0.385	0.086
VQAScore	0.225	0.641	-4.906	18	< 0.001	< 0.001	-0.267
CLIPScore	0.055	0.817	-2.113	18	0.024	0.049	-1.485

significantly outperformed GPT-4o in both VQA and CLIP-Scores, with two-sided p being < 0.001 and 0.049, respectively, indicating superior semantic understanding and textual alignment. In terms of aesthetic score, GPT-4o had a slightly higher mean of 5.09 compared to SDXL of 5.01, however, this difference was not statistically significant with two-sided p of 0.385, indicating that both models are perceived as similarly appealing in terms of visual quality.

TABLE VII
MEANS: LPIPS TO HUMAN (GPT-4O VS SDXL)

Model	Mean
LPIPS to Human (GPT-4o)	0.495
LPIPS to Human (SDXL)	0.5178

A paired samples t-test was conducted to compare the perceptual similarity (LPIPS) of GPT-4o and SDXL outputs to human-designed emojis, shown in Table VIII.

TABLE VIII
PAIRED SAMPLES T-TEST: LPIPS TO HUMAN (GPT-4O VS SDXL)

Comparison	t	df	One-sided p	Two-sided p
GPT-4o - SDXL	-1.432	29	0.081	0.163

Interestingly, GPT-4o scored lower on the LPIPS metric with a mean of 0.495 compared to 0.518, suggesting its outputs were closer in perceptual similarity to human emojis, however, this result was not statistically significant with a two-sided p of 0.163.

VII. HUMAN EVALUATION

Paired sample t-tests were done on the Turing-style test as seen in Table X and Table XI

TABLE IX
MEAN AI GUESS ACCURACY IN A TURING-STYLED SURVEY

Guess Type	Mean (%)
Total Correct AI Guesses	69
Total Wrong AI Guesses	31

Participants correctly identified AI-generated emojis significantly more than chance, with a mean accuracy of 69% and a two-sided p of < 0.001. This suggests that AI outputs are not yet fully indistinguishable from human-made ones. An interesting pattern to note when evaluating the individual answers is that most participants were able to fully identify the SDXL emojis; however, struggled with identifying the GPT-4o emojis. Table XI shows that all emotion categories were statistically significant in being identified, with higher rates observed for confused, embarrassed, and smiling emojis, suggesting that certain AI-generated emotional expressions are more easily distinguishable.

TABLE X
PAIRED SAMPLES T-TEST: ACCURACY OF AI IDENTIFICATION IN TURING-STYLED SURVEY.

Comparison	t	df	One-sided p	Two-sided p	Mean Diff. (%)
Correct vs. Wrong AI Guesses	5.203	24	< 0.001	< 0.001	38

TABLE XI
EMOTION-SPECIFIC ACCURACY IN AI GUESS IDENTIFICATION (TURING-STYLED SURVEY). MEAN VALUES REFLECT CORRECT GUESSES OUT OF 2 TRIALS PER EMOTION.

Emotion	% Correct AI Guesses (Mean)	t	One-Sided p	Two-Sided p
Angry	1.32 / 2	3.361	0.001	0.003
Surprised	1.24 / 2	2.753	0.006	0.011
Fearful	1.36 / 2	3.674	< 0.001	0.001
Disgust	1.28 / 2	3.055	0.003	0.005
Love	1.28 / 2	3.055	0.003	0.005
Confused	1.52 / 2	5.099	< 0.001	< 0.001
Proud	1.36 / 2	3.674	< 0.001	0.001
Embarrassed	1.56 / 2	5.527	< 0.001	< 0.001
Smiling	1.56 / 2	4.802	< 0.001	< 0.001
Sad	1.32 / 2	3.361	0.001	0.003

VIII. VALIDITY, RELIABILITY AND GENERALIZABILITY

The study demonstrates strong internal and construct validity through objective computational metrics. However, ecological and cultural validity are limited, as evaluations lacked human input and did not consider cross-cultural emotion interpretation. Reliability is supported by standardised workflows and repeatable scoring. However, generalizability remains a key limitation due to the small sample size, narrow emotional range, and lack of diverse user perspectives.

IX. COMPARISON OF RESULTS

The results of this study, which evaluated AI-generated emojis across emotional accuracy, aesthetic quality, and stylistic consistency, showed both alignment and divergence from existing research in the field. In contrast to earlier studies, this project adopted a comprehensive, metrics-driven evaluation framework, offering a more objective assessment of emotional and visual fidelity. The emotional accuracy of generated emojis in this study was quantified using CLIPScore and VQAScore, offering a scalable and replicable method of determining emotional alignment. This contrasts with [10], which used human surveys to assess emotional accuracy, offering higher internal validity but less scalability. The lack of human validation in this study may introduce model bias and limit ecological validity [10] also reported meant LPIPS scores for emojis generated by different GANs; EAE-GAN (0.30), StyleGAN (0.28), Style-ALAE (0.18), and PCGAN (0.13). In comparison, GPT-4o and SDXL had scores of 0.297 and 0.428, respectively, demonstrating that SDXL has the highest diversity of outputs among those mentioned. However, whether this is beneficial depends on design goals: a higher LPIPS score may be preferred for creative flexibility and emotional nuance, while lower LPIPS score may be better for maintaining a consistent art style across emojis. Similarly, [8] demonstrated that AI-generated faces could be perceived as more trustworthy than real ones, aligning conceptually with the hypothesis of this paper and supporting the idea that AI can rival human designs in emotional clarity. However, [8] leveraged human input and insight, which gives the results higher ecological validity.

X. Conclusion

This study explored whether GAI models, such as SDXL and GPT-4o, can generate emotionally expressive emojis comparable to human-designed ones in terms of emotional accuracy, aesthetic appeal, and stylistic consistency. The findings support the hypothesis that *AI-generated emojis can convey intended emotions at a similar level to human-designed emojis*, showing significant promise and room for improvement. In response to RQ1, the results showed that AI-generated emojis demonstrated comparable emotional accuracy, with SDXL achieving the highest VQAScore of 0.816 and CLIPScore of 30.692, outperforming both human-designed sets and GPT-4o. On the other hand, results from a Turing-style survey revealed that participants were able to correctly identify AI-generated emojis 69% of the time, with statistical significance, though GPT-4o-generated emojis were harder to distinguish, suggesting a closer visual resemblance to human-created designs.

For the RQ2, the AI-generated emojis performed significantly higher in aesthetic score, suggesting that GAI can produce visually pleasing designs. In terms of stylistic consistency, GPT-4o had the second lowest LPIPS score of 0.297, with SDXL having the largest LPIPS score of 0.428. This suggests that GPT-4o-generated emojis maintained a more cohesive visual style across different expressions, while SDXL produced more varied outputs, potentially useful in cases demanding greater expressiveness or diversity. When comparing the stylistic consistency in comparison to the human-designed sets, a mean LPIPS score of 0.495 and 0.517 was calculated for GPT-4o and SDXL, suggesting that the outputs generated by GPT-4o are closer to human emojis, however, this result was not statistically different.

Despite promising results, the research has limitations. The minimal amount of qualitative user input on emotional accuracy reduces ecological validity. Moreover, cultural bias and generalizability are limited due to the small emoji set, platform styles evaluated, and not considering emoji meanings in different cultures.

Future research should include diverse human feedback to better validate and contextualise the emotional accuracy of AI-generated emojis beyond computational metrics. Expanding the sample size and evaluating the emojis in a real-world communication setting would provide a more comprehensive understanding of their effectiveness. Additionally, the study could explore the impact of creating custom emojis from user prompts, tailoring visual communication to individual needs and preferences.

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